Gildardo Martinez

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Air Force Veteran, U.S. Secret Clearance (Expired)

SUMMARY

Equipment engineer with hands-on dry etch experience across Applied Materials Centura (DPS Metal Gate, 5200 DPS, Super-E, eMax 300mm), LAM Research Rainbow TCP (9600), Tokyo Electron Unity and Telius, SPTS Omega FXP, Synapse, and Rapier DSIE, and Axcelis Fusion ES asher. Resolved chronic wafer sticking on the LAM 9600 by diagnosing a missing dechuck step in the recipe and verifying the correction across affected processes. B.S. Physics, U.S. Air Force Aircraft Metals Technologist (Secret-eligible).

EDUCATION

University of San Diego **|** BS in Physics — January 2025
San Diego Miramar College **|** AA in Mathematics — August 2022
Community College of the Air Force **|** AS in Aircraft Metals Technology — May 2017

ENGINEERING PROJECTS AND EXPERIENCE

Equipment Engineer | Skorpios Technologies, Inc. | Temecula, CA — May 2025 - Present

- Sustain, optimize, and improve equipment performance and reliability across a dry etch fleet that includes Applied Materials Centura (DPS II Metal Gate, 5200 DPS, Super-E, eMax 300mm), LAM Research Rainbow 9600 TCP, Tokyo Electron Unity and Telius, SPTS Omega FXP, Synapse, and Rapier DSIE, and Axcelis Fusion ES.
- Diagnosed and resolved chronic wafer sticking on the LAM Rainbow 9600 by identifying a missing dechuck step in the recipe, then implemented and verified the correction across affected processes; increased affected-process throughput from ~1 wafer/day to 36 wafers/day.
- Perform hands-on component-level troubleshooting across RF/plasma subsystems, electrostatic chuck behavior, helium backside cooling, vacuum integrity, robotics and sensors, control circuits, and process gas delivery.
- Drive Pareto-led root cause and corrective action on repeat down events across the dry etch fleet, converting recurring issues into permanent fixes and updates to PMs and SOPs.
- Execute scheduled and unscheduled preventive maintenance, support tool start-up and qualification, and author PM and recovery documentation aligned with ISO requirements.
- Track equipment performance using availability, MTBF, MTTR, and PM compliance, and translate trends into prioritized improvement actions with ROI justification.
- Coordinate with vendor field service engineers and parts sourcing to minimize downtime, and partner with Process, Production, Facilities, and Procurement groups on cross-functional improvement work.

Research Assistant | University of San Diego | San Diego, CA — January 2023 - December 2024

- Engineered a Fourier Light Field Microscope for 3D imaging, integrating optics and computation for biomedical data capture.
- Created algorithms to analyze and measure particle dynamics in 3D, contributing to advanced microscopy techniques.
- First-authored peer-reviewed publication in Soft Matter and presented findings at APS 2024.

Aircraft Metals Technologist | United States Air Force | Whiteman AFB — February 2015 - December 2018

- Modeled components in SolidWorks and generated CNC tool paths using FeatureCAM.
- Machined structural aircraft components on CNC lathes and mills to tight tolerances.
- Interpreted technical drawings and engineering specifications to fabricate mission-critical parts.
- Conducted inspection and validation to maintain airworthiness standards.
- Supported operational readiness in high-reliability aerospace environments.

SKILLS

Dry Etch Toolsets: AMAT Centura (DPS, Super-E, eMax), LAM Rainbow TCP (9600), TEL Unity and Telius, SPTS (Omega FXP, Synapse, Rapier DSIE), Axcelis Fusion ES asher, Mattson Aspen III asher

Methods: structured root cause, Pareto-led repeat-failure elimination, SPC, scheduled and unscheduled PM, tool start-up and qualification, ISO compliance, MTBF / MTTR /availability tracking, vendor and FSE coordination

Foundations: B.S. Physics, USAF Aircraft Metals Technologist (Secret-eligible), Python, SolidWorks, GD&T